WebXSkill: Skill Learning for Autonomous Web Agents

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Paper version: arXiv 2604.13318v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Textual skills cannot be executed; code skills execute without step-level guidance.*

The paper calls this a **grounding gap** and pairs an action program with per-step natural-language guidance.

Motivation: three gaps in prior skills

1. **Textual workflows** guide, but need re-grounding.
2. **Code skills** run as black boxes; no recovery.
3. **Flat libraries** ignore the current page.

Evaluation setup

WebArena · 154 tasks | WebVoyager · 11 sites | Online-Mind2Web · 300 | GPT-5 | Qwen3.5-122B-A10B | 30-step budget | 2,500 synthetic tasks | 3 judges (OM2W)

- **All online**; no offline step accuracy.
- **Not held out:** skills built on the test sites.

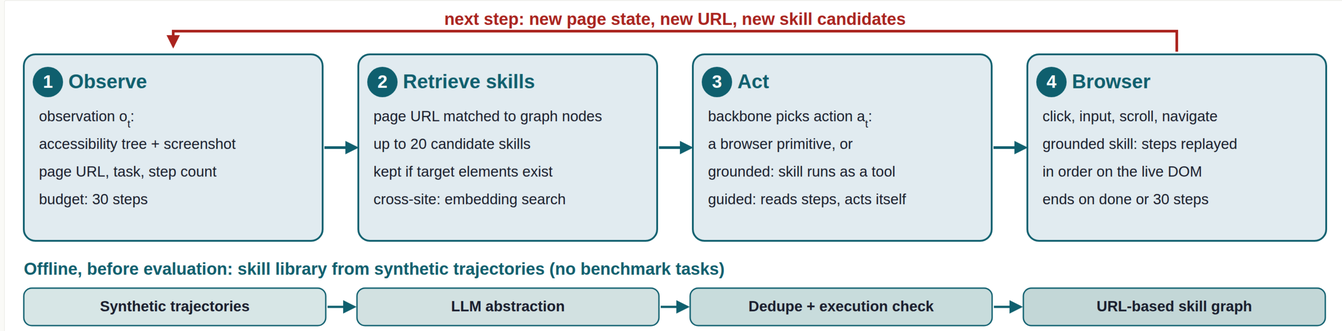
Our read (CUA Radar)

81	★★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; same 30-step budget; strong baselines; step counts; component ablation; code.
- **Flags:** AP5 (skills built on test sites); no seeds or variance; judge not validated.
- Pattern C19. Transfer: WebArena skills on WebVoyager. Authors' numbers, not rerun.

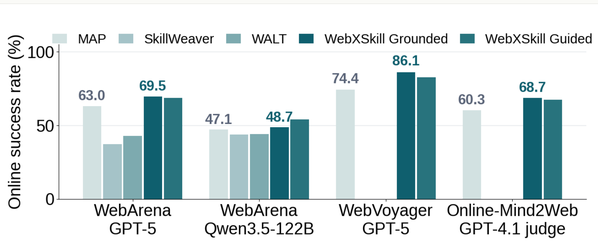
Method: extract, index by URL, deploy in two modes

- **Sees / acts:** accessibility tree + screenshot in; browser actions or a skill call out (web, browser-use harness).
- **Skill:** typed parameters, element-bound steps, per-step guidance; run once to verify.
- **Signal:** LLM-abstracted synthetic trajectories (2,500 WebArena tasks); skills kept only if they run; no fine-tuning described. Skills can also be exposed as a readable plan or as a tool.



Our redraw (not the paper's figure): the web-agent loop with URL-graph retrieval, and the offline skill build.

Key results: online success rate (%)

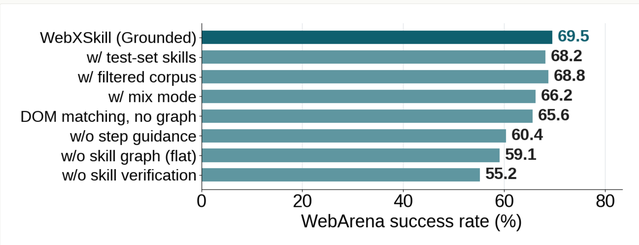


Tables 2–4, redrawn: GPT-5 unless noted; Online-Mind2Web with the GPT-4.1 judge.

WebArena (GPT-5): grounded 69.5, guided 68.8, MAP 63.0

SO WHAT → *Qwen3.5-122B favors guided (53.9). SkillWeaver and WALT were not run on live sites: their skills are not released for them (paper).*

Ablation and retrieval: WebArena success (%)



Tables 9–10, redrawn: GPT-5, grounded mode; one change at a time.

SO WHAT → *Removing **skill verification** costs most. The URL graph beats DOM matching and a flat library, and step guidance adds several points. Test-set skills and a corpus filtered of near-matching sources score close to the full run; the paper reads this as procedures that generalize across task wording. Mix mode ends up choosing grounded.*

Headline Numbers

69.5%
WebArena, GPT-5
vs 59.7% Vanilla, 63.0% MAP

- **86.1%** WebVoyager
- **85.1%** WebVoyager, WA skills
- **53.9%** Qwen guided

Controls: WebArena, GPT-5

CONTROL	WEBXSKILL	COMPARED WITH
Graph/DOM/flat	69.5/65.6/59.1	same library
Test-set skills	68.2	synthetic 69.5
SW / WALT skills in ours	59.7 / 62.3	own 37.4 / 42.9
WebArena skills on WebVoyager	85.1	MAP 74.4

No seeds or std reported; 154 WebArena tasks

Takeaway

Step-guided executable skills raise online success; verification and graph retrieval matter.

591 skills | usage rate 70.8%

Scan to Read

Paper | Code